ChatGPT

Minimising total treatment time

- Using the given code, one finds

$$f(\infty) = \sum_{k=3}^{\infty} (1/2)^k + (\text{tiny corrections}) = 1/4,$$

  since $1/2+1/4+1/8+\cdots=1$ [1]. Thus $a_2=2f(\infty)+\tfrac12=2\cdot\tfrac14+\tfrac12=1$ (up to negligible terms).

- Hence $a_1=1\ge a_2\approx1$, so the fastest infusion is therapy 1. By linear programming (a linear objective on the line segment) the minimum is at an endpoint [2]. Concretely, the minimum time is achieved at $(x_1,x_2)=(1,0)$ (giving total dose 1).

- The infinity-norm distance from $(0.1,0.9)$ to $(1,0)$ is $\max(|0.1-1|,|0.9-0|)=0.9$, which is much larger than 0.15. Therefore $(0.1,0.9)$ is **not** within 0.15 of the minimiser.

Answer: No, $(0.1,0.9)$ lies 0.9 away (in $|\cdot|_\infty$) from the optimal point (approximately $(1,0)$), so it is not within 0.15 of a minimiser [1] [2].

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[1] Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series

[2] linear programming - Simplex Algorithm: Why must the optimal value of the LP lie on the face or vertex of a polyhedron? - Computer Science Stack Exchange
https://cs.stackexchange.com/questions/84343/simplex-algorithm-why-must-the-optimal-value-of-the-lp-lie-on-the-face-or-verte